ANSWER KEY

Lab 8 VoIP call replay

Capture the traffic in Wireshark (Windows 11 PC, sign in with the password networkwalks) and follow the solution steps below.

Task 1. Capture the traffic — 10 marks

Sign in to the Windows PC and capture the network traffic. How many packets did Wireshark capture?

Answer: 1051

Solution: Sign in with the password networkwalks, double-click Wireshark on the desktop, then double-click the WiFi interface. The status bar shows Packets: 1051.

Task 2. Call setup protocol — 10 marks

A phone call was made on the network. Which protocol set up the call (INVITE, Ringing, OK)?

Answer: SIP

Solution: Filter

```
sip
```

. The Info column shows Request: INVITE, Status: 180 Ringing and Status: 200 OK. The setup protocol is SIP.

Task 3. Voice protocol — 10 marks

Which protocol carried the voice itself (the many small packets, one every 60 ms)?

Answer: RTP

Solution: Filter

```
rtp
```

. About 800 packets show RTP. RTP carries the voice.

Task 4. Caller extension — 10 marks

Which extension number made the call? Look at the SIP INVITE.

Answer: 101

Solution: Filter

```
sip.Method == "INVITE"
```

. The From line shows

```
sip:101@192.168.10.45
```

. The caller extension is 101.

Task 5. Caller name — 10 marks

What is the name of the caller?

Answer: Ali Raza

Solution: The INVITE says

```
From: "Ali Raza" <sip:101@192.168.10.45>
```

. The caller is Ali Raza.

Task 6. The codec

10 marks

Which codec carried the voice? Click one RTP packet and look at the payload type.

Answer: PCMU

Solution: In an RTP packet the line says Payload type: ITU-T G.711 PCMU (0). The codec is PCMU.

Task 7. Who answered

10 marks

Who answered the call? Play the call and listen to the first words.

Answer: Sara

Solution: The woman answers: “Good morning, Payments desk. Sara speaking.” The answer is Sara. The To line of the INVITE also says “Sara”.

Task 8. Card number

10 marks

The caller read out a card number to pay an invoice. What are the last 4 digits?

Answer: 2394

Solution: The man says the card number 4213 5588 7710 2394. The last 4 digits are 2394.

Task 9. Expiry date

10 marks

What expiry date did the caller say?

Answer: 09/28

Solution: The man says “Zero nine, twenty eight.” The expiry date is 09/28.

Task 10. The fix

10 marks

How can the company stop people from listening to calls like this?

Answer: Encrypt the call (SRTP)

Solution: Calls must be encrypted. SRTP encrypts the voice packets, and TLS protects the SIP messages. The answer is Encrypt the call (SRTP).

SUMMARY

Call setup:	SIP (UDP port 5060)
Voice:	RTP, G.711 PCMU
Caller:	Ali Raza, extension 101
Answered by:	Sara (extension 202)
Card:	ends 2394, expiry 09/28
Fix:	SRTP (encrypted calls)

Watch the Wireshark video playlist: https://www.youtube.com/playlist?list=PLG-wUGWU39ro